

Cosmic Microwave Background

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Almost every person linked directly or indirectly to Academia has heard the name of *Cosmic Microwave Background* but not everyone knows what *exactly* is this. Here, I will give a brief introduction of this deep topic which is a part of Cosmology, the study of history of the universe.

All of this started with two engineers, Arno Penzias and Robert Wilson, from Bell Laboratories experimenting with a supersensitive horn antenna, originally built to detect radio waves. They were looking for sources that might contribute to noise in their readings so that they could remove them to measure faint radio waves. The optimum frequency chosen for the measurement was 4080 MHz and a total temperature of $6.7K$ was recorded as noise. Out of this, the ohmic losses came out to be $0.8 \pm 0.4K$ while the backlobe response from the ground was about $0.1K$. The atmospheric contribution due to absorption was estimated to be about $2.3 \pm 0.3K$. They still had a contribution of $3.5 \pm 0.1K$ which was unexplained. They did many observations at different elevation angles in different seasons of the year but this part of the noise was consistent. They finally concluded that there was something in the sky which acted as a source of radio waves and was isotropic, unpolarized and independent of seasonal variations. (1)

You should ask “Why is the noise represented in the units of temperature and is there any relation between the wavelength of radio waves that the antenna measures and the temperature readings we have here?”

The answer to the first question is that it’s a thing that the physicists have agreed on. Another very famous example is $c = \hbar = 1$. The answer to the second question is, Yes, we have a such a relation, i.e.,

$$\lambda_{max} = \frac{b}{T}$$

where λ_{max} is the wavelength at peak emission and T being the temperature corresponding to it. b , here, is a constant $\approx 2.89777m.K$ and the relation is known as *Wein’s Displacement Law*. Also, if you are curious about the exact dimensions of the horn antenna, you can refer to (1). Figure 1 shows the antenna.

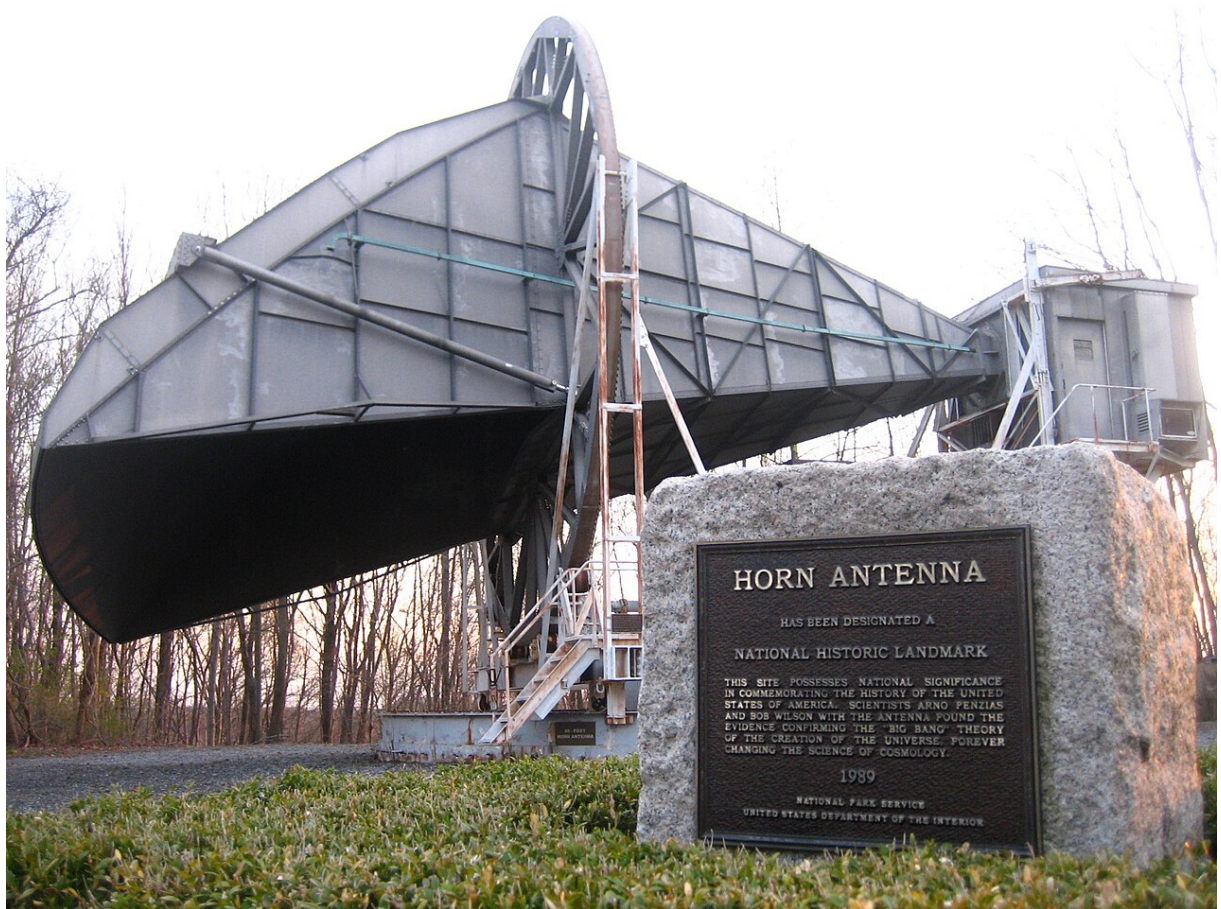


Figure 1: The Horn Antenna (2)

Now, there were many theories being developed to explain the origin and fate of this universe. The most important were,

1. The assumption of continuous creation which avoids the singularity by postulating a universe expanding for all time and continuous but slow creation of new matter in the universe.
2. The assumption that creation of new matter is intimately related to the existence of singularity.
3. The assumption that the singularity results from mathematical over-idealization, the requirement of strict isotropy and uniformity and that it would not occur in the real world.

If this third premise is accepted tentatively as a working hypothesis, it carries with it a possible resolution of a paradox (of conservation of baryons and leptons), for the matter we see about us now may represent the same baryon content of the previous expansion of a closed universe, oscillating for all time. In this picture it is essential to suppose that at the time of maximum

collapse the temperature of the universe would exceed $10^{10} K$, in order that the ashes of the previous cycle would have been reprocessed back to the hydrogen required for the stars in the next cycle.

In the early stages, almost instantaneous equilibrium between the electron pair-production was achieved due to large interactions, and the universe achieved a thermal equilibrium distribution (like that as depicted in Figure 2). The equilibrium lead to a temperature being associated with it and now as the universe went under an expansion, it adiabatically cooled which more or less preserved the overall thermal character of the distribution. You can understand this by visualizing one of the four graphs in Figure 2 (i.e. at four different temperatures) being moved to the right, without changing its shape. (Can you think why we move it to the right?)

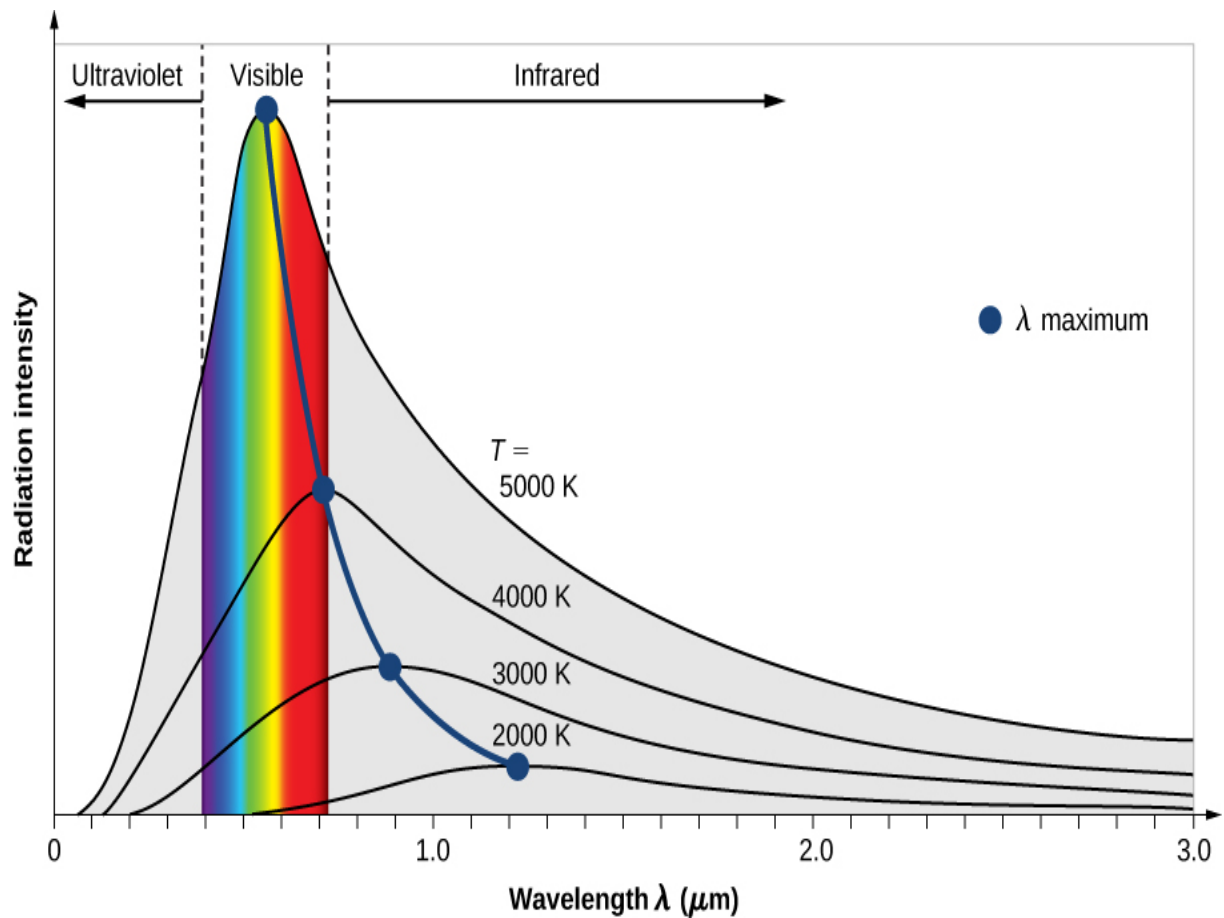


Figure 2: Black Body Spectrum (3)

Now, we have another problem, the Density Problem. To avoid a catastrophic helium production, either the present matter density should be $3 \times 10^{32} \text{ gm/cm}^3$, or there should exist some form of energy content with very high pressure, such as the zero-mass scalar, capable of speeding the universe through the period of helium formation. This zero-mass scalar field is what is known as Inflation and is responsible for the exponential growth of the universe. There were other Density problems as well related to the curvature of the universe, i.e., whether the universe is closed or open but we won't be discussing it here.

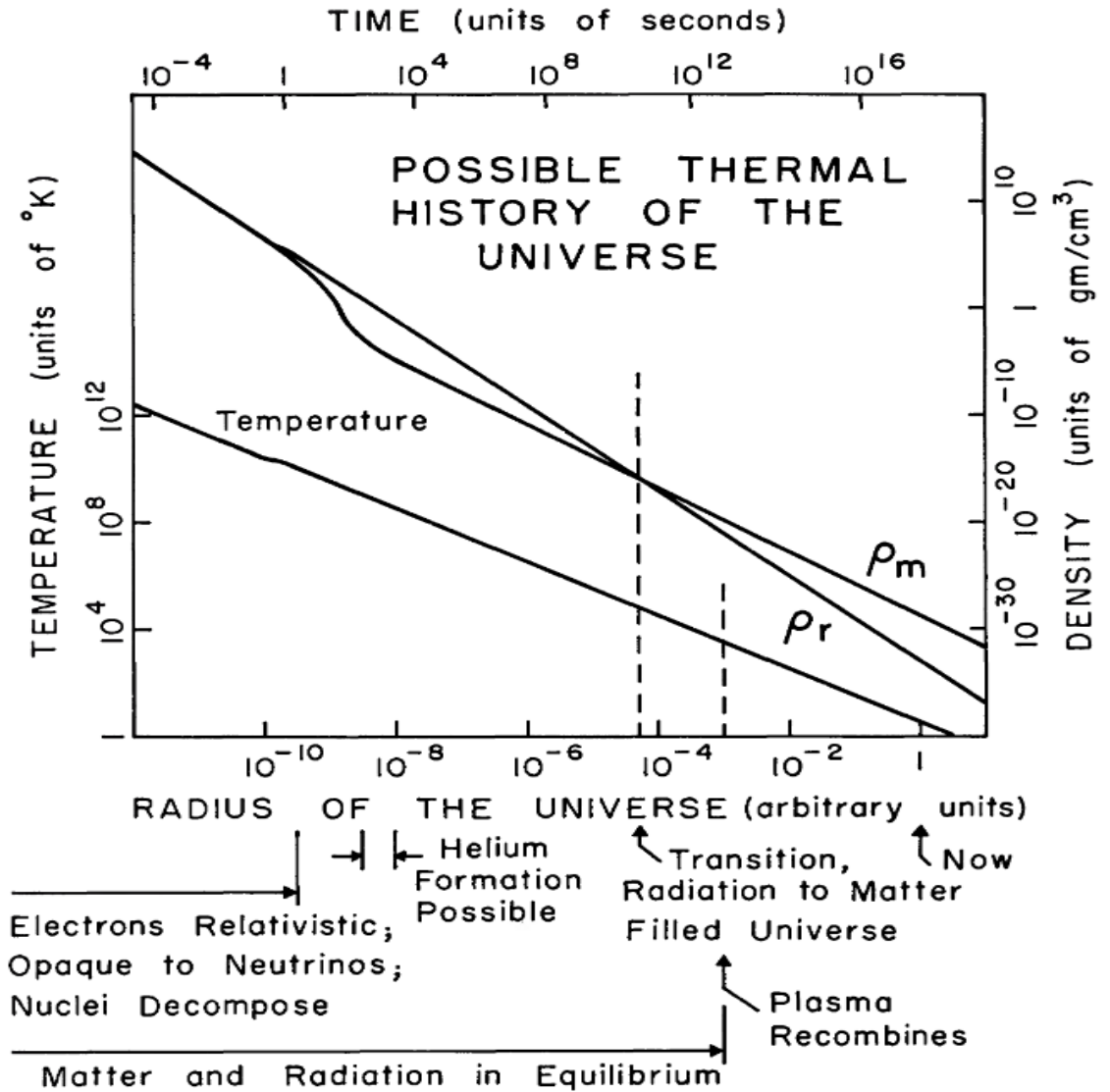


Figure 3: Sequential order of the major events (4)

The images above and below represent the events that happened. Most of the events in the above picture are relatively closer to the big bang than the below one.

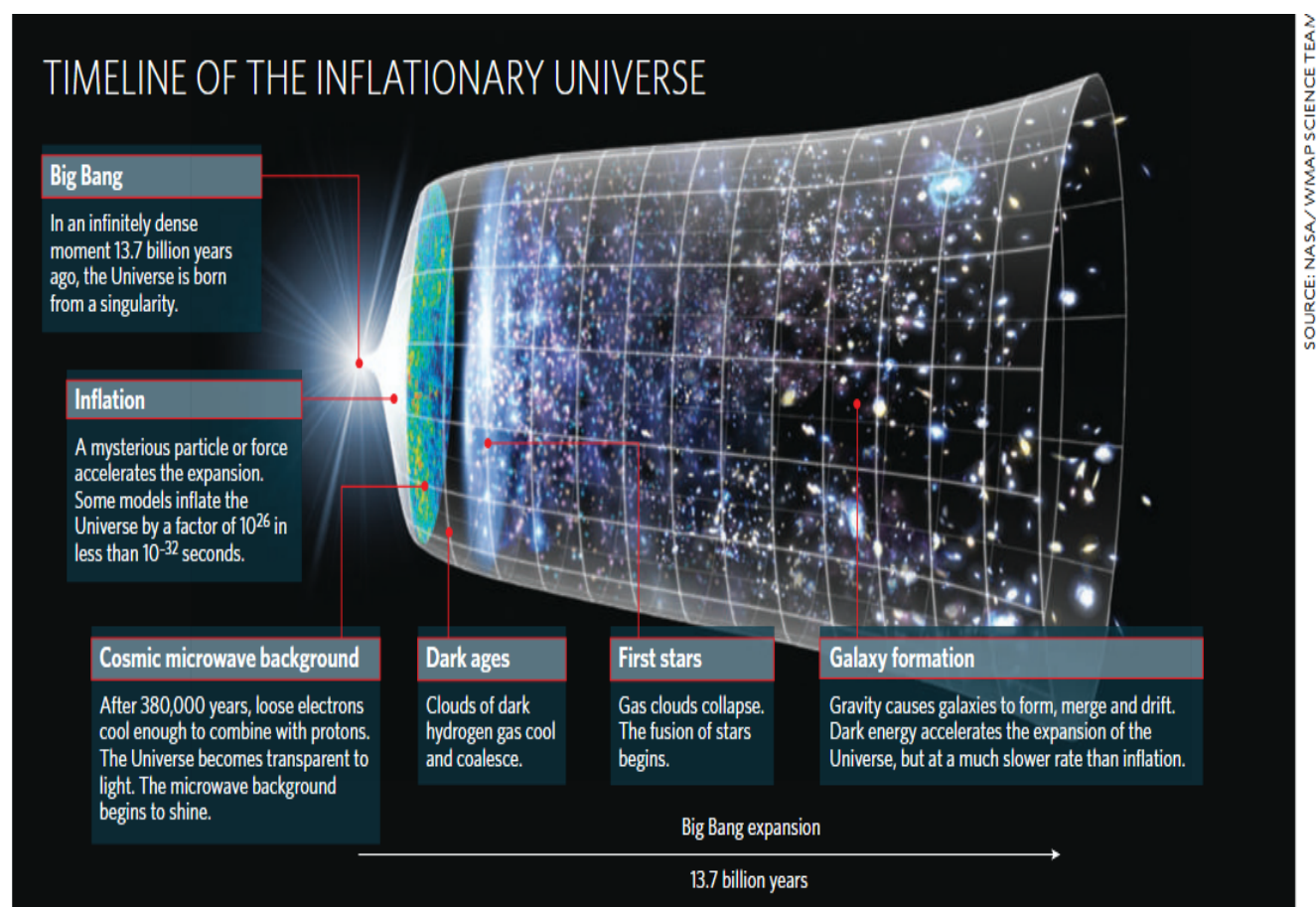


Figure 4: Big Bang and after (5)

Note that in Figure 3, the helium formation refers to the formation of helium nuclei and not the atoms. As the recombination phase (when free protons combined with free electrons to form hydrogen atom) happened, photons were released (due to this recombination) which were highly red-shifted ($z \approx 1100$) to form the CMB. Before the recombination, the universe was a hot, dense plasma where photons were constantly scattered by charged particles known as Thompson Scattering. (4)

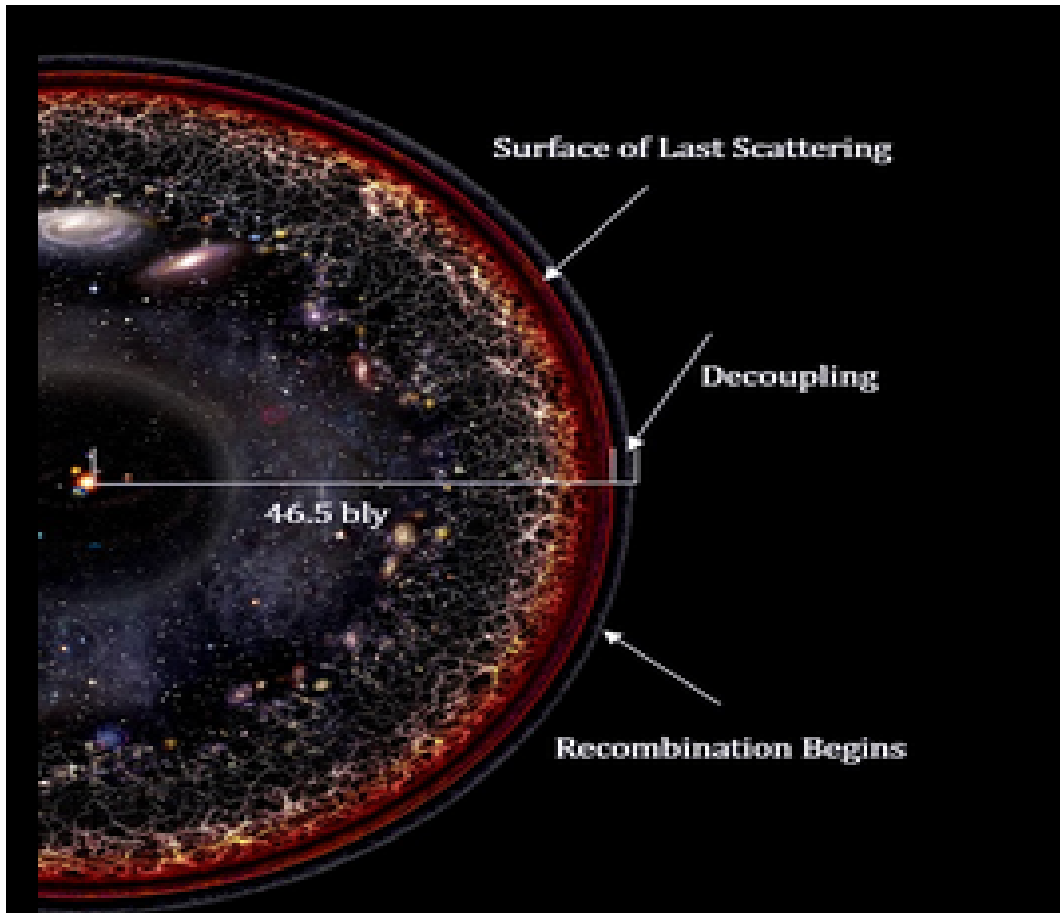


Figure 5: The Observable Universe (6)

As can be seen in Figure 5, the surface of the last scattering refers to the point when the scattering stopped and we are able to photons only after that point. Before the recombination, free electrons acted as a glue between photons and baryons which made the cosmological plasma a tightly coupled photon-baryon fluid. The Decoupling in Figure 5 refers to the decoupling of photons from the baryonic matter. The preceding line is actually linked with a relatively complex topic and maybe skipped without the loss of continuity. Now observe Figure 6. This *is* the Cosmic Microwave Background. “Why are mainly three different colors?”, you should ask. Well, the colors represent different temperatures. “But you said the CMB was isotropic”. Yes, *roughly*. There are anisotropies of the order 10^{-5} . This is so minute that it took years, if not decades, to realize the anisotropies. So, what happened was, the inevitable quantum fluctuations, that were supposed to very small, rapidly grew due to even more rapid, exponential expansion of the universe, and this is what caused the small density variations. Now, we know that it takes energy to overcome a potential barrier. So when photons corresponding to the CMB were released, they had to overcome different potential barriers induced by different densities of space which shifted the frequency/energy of the photons differently, and thus we see these tiny temperature fluctuations.

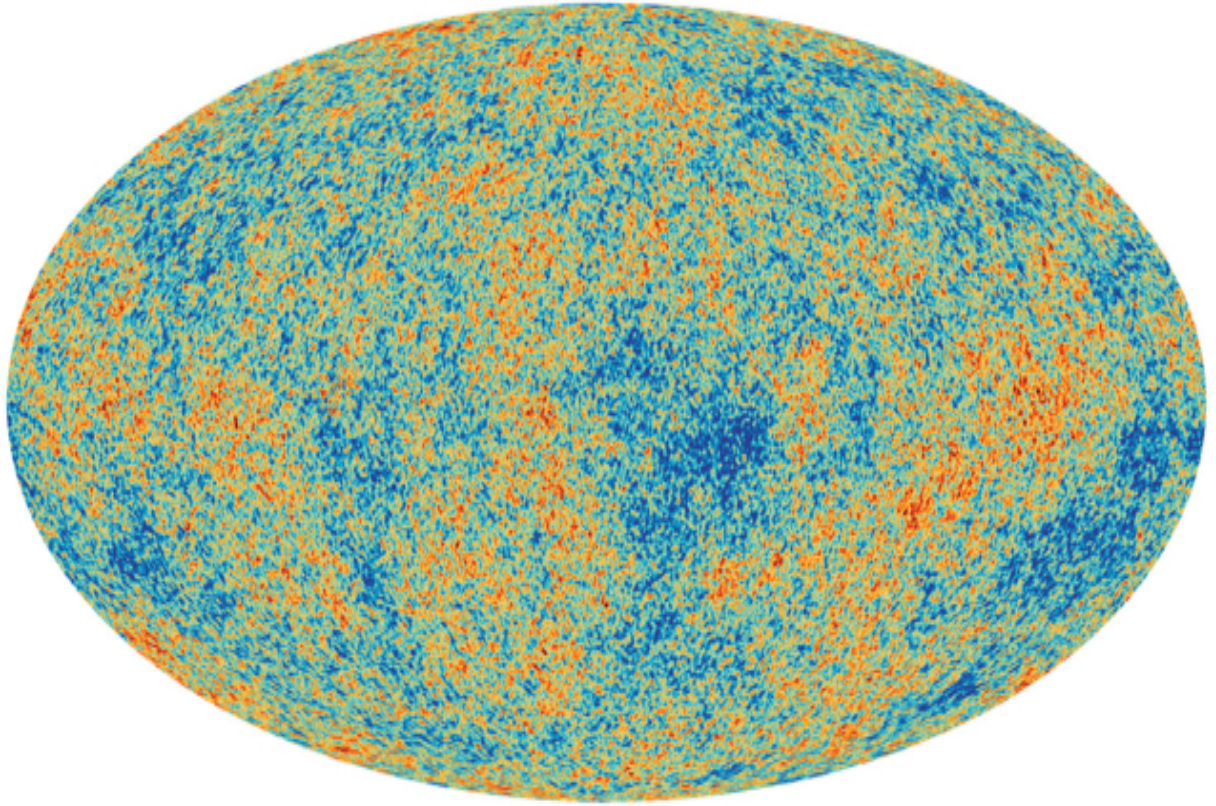


Figure 6: The Cosmic Microwave Background (7)

Figure 6 shows the temperature distribution of the universe mapped from Earth and neighbors (space based observatories) in microwave wavelength. The blue spot represent a colder region while the reddish represent the hotter ones with the temperature of yellow regions being somewhere in between.

The explanation in this article was very basic as compared to the complexities we deal with now. For example, the anisotropies are also influenced by something known as Acoustic Oscillations, but that topic happen to be much more advanced as compared to what we dealt with in this article. So, we will keep that discussion for future. The people familiar with the topic can refer to (8). The curious people who want to be haunted by equations can also refer to the same.

References

- [1] A. A. Penzias and R. W. Wilson, “A measurement of excess antenna temperature at 4080 mhz,” in *A Source Book in Astronomy and Astrophysics, 1900–1975*. Harvard University Press, 1979, pp. 873–876.
- [2] “Discovery of cosmic microwave background radiation - Wikipedia — en.wikipedia.org,” https://en.wikipedia.org/wiki/Discovery_of_cosmic_microwave_background_radiation, [Accessed 14-06-2025].
- [3] “Black-body radiation — mit.edu,” <https://www.mit.edu/~ashrstnv/black-body-radiation.html>, [Accessed 14-06-2025].
- [4] R. H. Dicke, P. J. E. Peebles, P. G. Roll, and D. T. Wilkinson, “Cosmic black-body radiation.” *Astrophysical Journal*, vol. 142, p. 414-419, vol. 142, pp. 414–419, 1965.
- [5] E. Hand, “Cosmology: The test of inflation,” *Nature*, vol. 458, no. 7240, pp. 820–824, Apr. 2009.
- [6] “- YouTube — google.com,” <https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3Dz26hH0nhpuw&psig=AOvVaw1JacjdAkdAy4Mr2G83S7W1&ust=1749984938143000&source=images&cd=vfe&opi=89978449&ved=0CBkQ3YkBahcKEwig84a54PCNAxUAAAAAHQAAAAAQBA>, [Accessed 14-06-2025].
- [7] cern, “Exploring the CMB like never before – CERN Courier — cerncourier.com,” <https://cerncourier.com/a/exploring-the-cmb-like-never-before/>, [Accessed 14-06-2025].
- [8] W. Hu and S. Dodelson, “Cosmic microwave background anisotropies,” *Annual Review of Astronomy and Astrophysics*, vol. 40, no. 1, pp. 171–216, 2002.